

Lecture 6 continue

① The EKF summary for the VINS

② Observability analysis

The EKF — VINS: IMU + vision

The state vector: $x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} \frac{q}{2}\theta & 3 \times 1 \\ q_p & 3 \times 1 \\ q_{v_2} & 3 \times 1 \\ b_g & 3 \times 1 \\ b_a & 3 \times 1 \\ \underline{q_{pf}} & 3 \times 1 \end{bmatrix}$ 15 DOF for x_1 , 2 IMU state
 $SO(3) + R^3 + R^3 + R^3 + R^3$
 3 for feature, 1 3D feature

The Error state for $x = \hat{x} \oplus \tilde{x}$

$$\tilde{x} = \begin{bmatrix} \delta\theta \\ \tilde{q}_p \\ \tilde{q}_{v_2} \\ \tilde{b}_g \\ \tilde{b}_a \\ \tilde{q}_{pf} \end{bmatrix} \quad \hat{x} = \begin{bmatrix} \frac{q}{2}\hat{\theta} \\ \hat{q}_p \\ \hat{q}_{v_2} \\ \hat{b}_g \\ \hat{b}_a \\ \hat{q}_{pf} \end{bmatrix}$$

$$\frac{q}{2}R = \frac{q}{2}\hat{R} \cdot \delta R \triangleq \text{Exp}(\frac{q}{2}\hat{\theta}) \cdot \text{Exp}(\delta\theta) \quad (1)$$

$$q_p = \hat{q}_p + \tilde{q}_p$$

$$q_{v_2} = \hat{q}_{v_2} + \tilde{q}_{v_2}$$

$$b_g = \hat{b}_g + \tilde{b}_g$$

$$b_a = \hat{b}_a + \tilde{b}_a$$

$$q_{pf} = \hat{q}_{pf} + \tilde{q}_{pf}$$

$$(1) \quad \frac{q}{2}R = \text{Exp}(\frac{q}{2}\theta), \quad \frac{q}{2}R = \frac{q}{2}\hat{R} \cdot \delta R \triangleq \text{Exp}(\frac{q}{2}\hat{\theta}) \cdot \text{Exp}(\delta\theta)$$

$$\text{Hence: } \text{Exp}(\frac{q}{2}\theta) = \text{Exp}(\frac{q}{2}\hat{\theta}) \cdot \text{Exp}(\delta\theta) \Rightarrow \frac{q}{2}\theta = \frac{q}{2}\hat{\theta} + J_r(\frac{q}{2}\hat{\theta}) \cdot \delta\theta$$

$$x = \hat{x} \oplus \tilde{x} \Rightarrow \begin{bmatrix} \frac{q}{2}\theta \\ q_p \\ q_{v_2} \\ b_g \\ b_a \\ q_{pf} \end{bmatrix} = \begin{bmatrix} \frac{q}{2}\hat{\theta} \\ \hat{q}_p \\ \hat{q}_{v_2} \\ \hat{b}_g \\ \hat{b}_a \\ \hat{q}_{pf} \end{bmatrix} \oplus \begin{bmatrix} \delta\theta \\ \tilde{q}_p \\ \tilde{q}_{v_2} \\ \tilde{b}_g \\ \tilde{b}_a \\ \tilde{q}_{pf} \end{bmatrix} = \begin{bmatrix} \frac{q}{2}\hat{\theta} + J_r(\frac{q}{2}\hat{\theta})\delta\theta \\ \hat{q}_p + \tilde{q}_p \\ \hat{q}_{v_2} + \tilde{q}_{v_2} \\ \hat{b}_g + \tilde{b}_g \\ \hat{b}_a + \tilde{b}_a \\ \hat{q}_{pf} + \tilde{q}_{pf} \end{bmatrix}$$

$$(2) \quad \frac{q}{2}\bar{q}, \quad \frac{q}{2}\bar{q} = \frac{q}{2}\hat{q} \otimes \delta q, \quad \delta q = \begin{bmatrix} 1 \\ \frac{1}{2}\delta\theta \end{bmatrix}, \text{ quaternion}$$

IMU dynamics: (from ...)

$$\begin{cases} w_m = w + b_g + n_g \\ a_m = a + b_a + n_a \end{cases}$$

Then:

$$\dot{g}_R = g_R \cdot [w_m - b_g - n_g]_x$$

$$g_{p_z} = g_{p_z}$$

$$g_{v_z} = g_R (a_m - b_a - n_a) + g_g, \quad g_g = \begin{bmatrix} 0 \\ 0 \\ -9.81 \end{bmatrix}$$

$$\dot{b}_g = n_{wg}$$

$$\dot{b}_a = n_{wa}$$

$$g'_{p_z} = 0_{3 \times 1}$$

$$\dot{x} = f(x, n_z)$$

$\Downarrow$

$$x_{k+1} = f_{\text{int}}(x_k, n_{z_k})$$

$f_{\text{int}}(\cdot)$: is the integration of $f(\cdot)$

Vision measurements (lecture 3)

$$z_c = h_c(x) + n_c$$

$$\begin{aligned} z_c &= \begin{bmatrix} c_x/c_z \\ y/c_z \end{bmatrix}, \quad c_p = \begin{bmatrix} c_x \\ c_y \\ c_z \end{bmatrix} = g_R^T (g_{p_f} - g_{p_z}) \\ &= (g_R^T z_c)^T (g_{p_f} - (g_R^T p_c + g_{p_z})) \\ &= z_c^T g_R^T (g_{p_f} - g_{p_z} - g_R^T p_c) \\ &= z_c^T g_R^T (g_{p_f} - g_{p_z}) - z_c^T z_c p_c \\ &= z_c^T g_R^T (g_{p_f} - g_{p_z}) + p_z \end{aligned}$$

EKF:- VINS:

Given: $x_k \sim \mathcal{N}(\hat{x}_k, P_k)$, prior information:

Propagation: $x_{k+1} = f_{\text{int}}(x_k, n_{z_k})$

$$\hat{x}_{k+1|k} = f_{\text{int}}(\hat{x}_k, 0)$$

$$\tilde{x}_{k+1} = \Phi_{f_{\text{int}}, b} \tilde{x}_k + G_k n_{z_k}$$

4/24/17

$$P_{k+1|k} = \mathcal{I}_{(k+1,k)} P_{k|k} \mathcal{I}_{(k+1,k)}^T + G_k R_{2d} G_k^T$$

where $\mathcal{I}_{(k+1,k)}$, G_k (from lecture 5), $\Xi_{1,2,3,4}$ are from lecture 5.

$$\mathcal{I}_{(k+1,k)} = \begin{bmatrix} \frac{G_k \hat{R}^T}{\hat{R}} \frac{G_k}{\hat{R}} & 0_3 & 0_3 & -\text{Tr}(\hat{R} \hat{R}^T) \delta t_k & 0_3 \\ -\left[\frac{G_k \hat{R}^T}{\hat{R}} \frac{G_k}{\hat{R}} - \frac{G_k \hat{R}^T}{\hat{R}} \frac{G_k}{\hat{R}} \delta t_k - \frac{1}{2} \frac{G_k \hat{R}^T}{\hat{R}} \frac{G_k}{\hat{R}} \delta t_k^2 \right] \frac{G_k}{\hat{R}} & I_3 & I_3 \delta t & \frac{G_k \hat{R}^T}{\hat{R}} \Xi_4 & -\frac{G_k \hat{R}^T}{\hat{R}} \Xi_2 \\ -\left[\frac{G_k \hat{R}^T}{\hat{R}} \frac{G_k}{\hat{R}} - \frac{G_k \hat{R}^T}{\hat{R}} \frac{G_k}{\hat{R}} \delta t_k \right] \frac{G_k}{\hat{R}} & 0_3 & I_3 & \frac{G_k \hat{R}^T}{\hat{R}} \Xi_3 & -\frac{G_k \hat{R}^T}{\hat{R}} \Xi_1 \\ 0_3 & 0_3 & 0_3 & I_3 & 0_3 \\ 0_3 & 0_3 & 0_3 & 0_3 & I_3 \end{bmatrix}$$

$$G_k = \begin{bmatrix} -\text{Tr}(\hat{R} \hat{R}^T) \delta t_k & 0_3 & 0_3 & 0_3 \\ \frac{G_k \hat{R}^T}{\hat{R}} \Xi_4 & -\frac{G_k \hat{R}^T}{\hat{R}} \Xi_2 & 0_3 & 0_3 \\ \frac{G_k \hat{R}^T}{\hat{R}} \Xi_3 & -\frac{G_k \hat{R}^T}{\hat{R}} \Xi_1 & 0_3 & 0_3 \\ 0_3 & 0_3 & I_3 \delta t_k & 0_3 \\ 0_3 & 0_3 & 0_3 & I_3 \delta t_k \end{bmatrix}$$

update: $z_c = h_c(x) + n_c$ $n_c \sim N(0, R)$

$$\hat{z}_c = h_c(\hat{x}_{k+1|k})$$

$$\tilde{z}_c = z_c - \hat{z}_c = z_c - h_c(\hat{x}_{k+1|k})$$

$$\tilde{z}_c = H \tilde{x} + n, \quad S_{k+1} = H P_{k+1|k} H^T + R_{k+1}$$

$$H = \frac{\partial \tilde{z}_c}{\partial \tilde{x}} \cdot \frac{\partial \tilde{x}}{\partial \tilde{p}_f} \quad \frac{\partial \tilde{z}_c}{\partial \tilde{p}_f} = \begin{bmatrix} \frac{1}{c_z} & 0 & -\frac{c_x}{c_z} \\ 0 & \frac{1}{c_z} & -\frac{c_y}{c_z} \end{bmatrix}$$

$$= \frac{\partial \tilde{z}_c}{\partial \tilde{p}_f} \left[\frac{1}{c_z} \hat{R}^T \left[\frac{G_k \hat{R}^T}{\hat{R}} \left(\frac{G_k \hat{R}^T}{\hat{R}} - \frac{G_k \hat{R}^T}{\hat{R}} \right) \right] \frac{1}{c_z} \quad -\frac{1}{c_z} \hat{R}^T \frac{G_k \hat{R}^T}{\hat{R}} \quad 0_3 \quad 0_3 \quad 0_3 \quad \frac{1}{c_z} \hat{R}^T \frac{G_k \hat{R}^T}{\hat{R}} \right]$$

Then: $K = P_{k+1|k} H^T S_{k+1}^{-1} = P_{k+1|k} H^T (H P_{k+1|k} H^T + R_{k+1})^{-1}$

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$$\hat{x}_{k+1|k} = \hat{x}_{k+1|k} + \tilde{x}_{k+1} = \hat{x}_{k+1|k} + K \tilde{z}_k$$

$$P_{k+1|k} = P_{k+1|k} - P_{k+1|k} H^T S_{k+1}^{-1} H P_{k+1|k}$$

The above is the basic idea for the EKF

- ① sliding window filter
- ② iterative EKF
- ③ square-root inverse filter
- ④ Square-root filter
- ⑤ Schmidt filter
- ⑥ invariant EKF
- ⑦ Equivariant filter

observability analysis:

observability describes whether the system can recover its initial states based on the all history measurements.

Assuming no noises, then:

$$t_0: \tilde{z}_1 = H_1 \tilde{x}_1$$

$$t_2: \tilde{x}_2 = \Phi_{(2,1)} \tilde{x}_1, \tilde{z}_2 = H_2 \tilde{x}_2 \Rightarrow \tilde{z}_2 = H_2 \Phi_{(2,1)} \tilde{x}_1$$

$$t_3: \tilde{x}_3 = \Phi_{(3,2)} \tilde{x}_2, \tilde{z}_3 = H_3 \tilde{x}_3 \Rightarrow \tilde{z}_3 = H_3 \Phi_{(3,2)} \Phi_{(2,1)} \tilde{x}_1 \triangleq \Phi_{(3,1)} \tilde{x}_1$$

.....

$$\dots \tilde{x}_n = \Phi_{(n,n-1)} \tilde{x}_{n-1} \dots \tilde{z}_n = H_n \Phi_{(n,n-1)} \dots \Phi_{(2,1)} \tilde{x}_1$$

$$t_k: \tilde{x}_k = \Phi(k, k-1) \tilde{x}_{k-1}, \quad \tilde{z}_k = H_k \tilde{x}_k \Rightarrow \tilde{z}_k = H_k \Phi(k, k-1) \dots \Phi(2, 1) \tilde{x}_1 = \Phi(k, 1) \tilde{x}_1$$

Hence:

$$\begin{bmatrix} \tilde{z}_1 \\ \tilde{z}_2 \\ \tilde{z}_3 \\ \vdots \\ \tilde{z}_k \end{bmatrix} = \underbrace{\begin{bmatrix} H_1 \\ H_2 \Phi(2, 1) \\ H_3 \Phi(3, 1) \\ \vdots \\ H_k \Phi(k, 1) \end{bmatrix}}_{\Theta} \tilde{x}_1$$

$$\Theta = \begin{bmatrix} H_1 \\ H_2 \Phi(2, 1) \\ H_3 \Phi(3, 1) \\ \vdots \\ H_k \Phi(k, 1) \end{bmatrix} \triangleq \begin{bmatrix} m_1 \\ m_2 \\ m_3 \\ \vdots \\ m_k \end{bmatrix}$$

observability matrix

let's study $m_k \triangleq H_k \Phi(k, 1)$

$$= \frac{\partial \tilde{z}_k}{\partial \hat{p}_f} \cdot \frac{\partial \tilde{z}_k^T}{\partial \hat{p}_f} \begin{bmatrix} \Gamma_1 & \Gamma_2 & \Gamma_3 & \Gamma_4 & \Gamma_5 & \Gamma_6 \end{bmatrix}$$

$$\Gamma_1 = \begin{bmatrix} L p_f - \hat{p}_{2,1} - \hat{v}_{2,1} \delta t_k - \frac{1}{2} \hat{g} \delta t_k^2 \end{bmatrix} \frac{\hat{g}}{2,1}$$

$$\Gamma_2 = -I_3 \quad \Gamma_3 = -I_3 \delta t_k$$

$$\Gamma_4 = \begin{bmatrix} L p_f - \hat{p}_{2,k} \end{bmatrix} \frac{\hat{g}}{2,k} \Phi_{1,4} - \Phi_{2,4} \quad \Gamma_5 = -\Phi_{2,5} \quad \Gamma_6 = I_3$$

we can easily find: N that $m_k \cdot N = 0$

$$N = \begin{bmatrix} \frac{\hat{g}}{2,1} \hat{g} & \vdots & 0_3 \\ -L p_{2,1} \hat{g} & \vdots & I_3 \\ -L \hat{v}_{2,1} \hat{g} & \vdots & 0_3 \\ 0_{3 \times 1} & \vdots & 0_3 \\ 0_{3 \times 1} & \vdots & 0_3 \\ -L p_{2,k} \hat{g} & \vdots & I_3 \end{bmatrix}$$

18x1 18x3

4 dof unobservable directions

{ 1 dof relates to global yaw
3 dof relates to global translation